

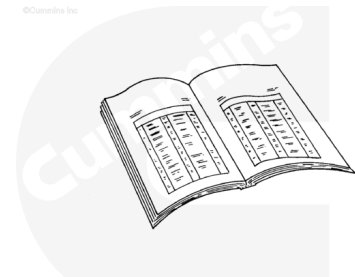
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Disconnect the batteries or air supply line to the air starter to prevent accidental engine starting. Refer to the OEM service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)

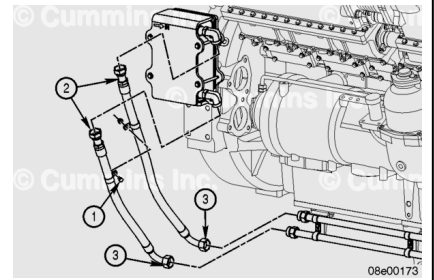
Remove

Note : Only remove the elbows from the marine gear oil cooler if necessary.

Remove the bolt securing the p-clips (1) together.

Remove the coolant line connectors (2) from the elbows on the marine gear oil cooler.

Remove the coolant line connectors (3) from the water transfer tube.

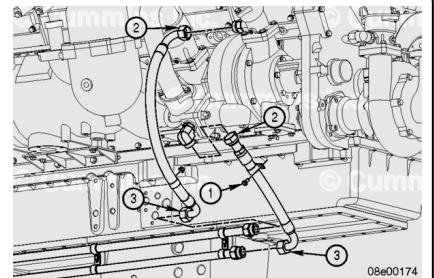


Note : Only remove the fittings from the LTA water pump if necessary.

Remove the bolt securing the p-clips (1) together.

Remove the coolant line connectors (2) from the fittings on the LTA water pump.

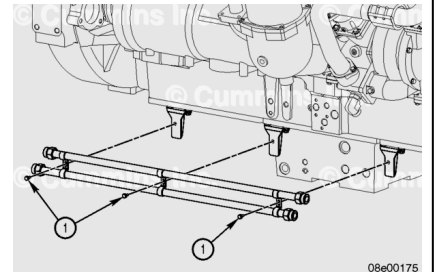
Remove the coolant line connectors (3) from the water transfer tube.



Note : There are 3 brackets on the QSK38 engine and 4 brackets on the QSK50 engine.

Remove the bolts (1), p-clips, and water transfer tubes.

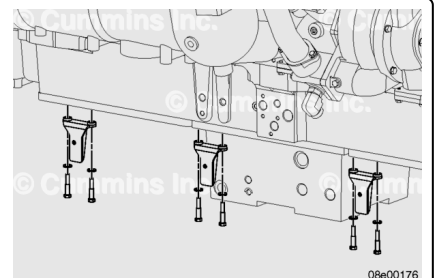
Discard the o-rings.



Note : If the lubricating oil pan is being removed, mark the location of the brackets to make sure they are installed in the same location.

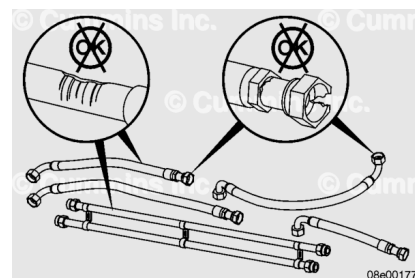
Note : There are 3 brackets on the QSK38 engine and 4 brackets on the QSK50 engine.

Remove the bolts, washers, and brackets from the oil pan.



Inspect for Reuse

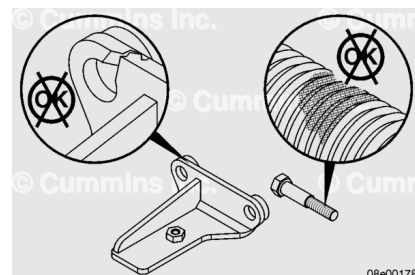
Inspect the coolant lines and water transfer tubes for cracks, fretting, or other damage. If damage is found, the part **must** be replaced.



Inspect the brackets and p-clips for cracks and other damage. Check that the coating on the water transfer tube clips has **not** been worn away.

Inspect the capscrews for damaged threads.

If damage is found, the part **must** be replaced.

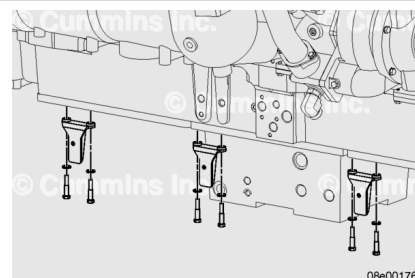


Install

Note : There are 3 brackets on the QSK38 engine and 4 brackets on the QSK50 engine.

Install the bolts, washers, and brackets into the lubricating oil pan.

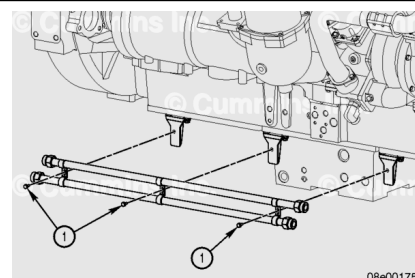
Torque Value: 45 n•m [33 ft-lb]



Note : Do not torque any fittings or bolts until all components have been installed to make sure of proper alignment.

Note : There are 3 brackets on the QSK38 engine and 4 brackets on the QSK50 engine.

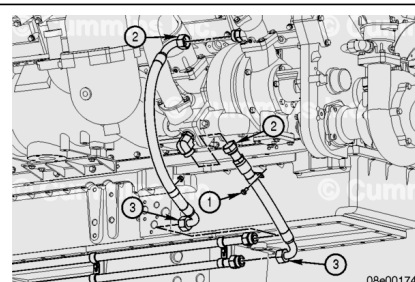
Install the bolts (1), p-clips, and water transfer tubes, using new o-rings.



If removed, install the elbows with new o-rings onto the engine and marine gear oil cooler.

Torque Value: 135 n•m [100 ft-lb]

Connect the coolant line connectors (2) to the fittings on the LTA water pump.



Connect the coolant line connectors (3) to the water transfer tubes.

Install the bolt securing the p-clips (1) together.

Connect the coolant line connectors (2) to the elbows on the marine gear cooler.

Connect the coolant line connectors (3) to the water transfer tubes.

Install the bolt securing the p-clips (1) together.

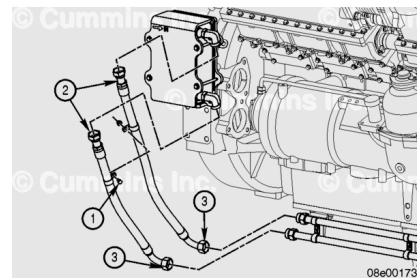
Tighten the hose fittings and clips.

Hose Fittings

Torque Value: 100 n•m [74 ft-lb]

Clips

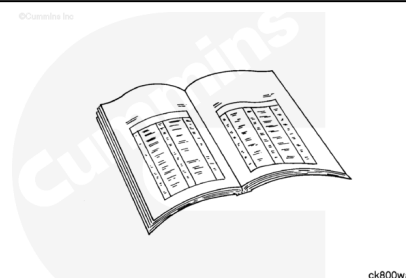
Torque Value: 25 n•m [221 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries or air starter air supply line. Refer to the OEM service manual.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Operate the engine and check for leaks

